

Measuring Local SLM Agents as Systems: Protocol Effects on Quality and Resource Use

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Local small-language-model (SLM) agent results are properties of a complete protocol, not of a model label alone. We test this claim with fixed local tool-call workloads from five fully imported BFCL-v3 single-turn category files. In a hash-pinned 2,480-case M4 control, jointly changing the declared request instruction and requested JSON shape moves exact match by up to 20.0 points on non-irrelevance tasks and reverses the ranking of two Qwen artifacts when abstention must be decided rather than supplied. A historical positive schema-order pattern is then tested prospectively over 6,400 attempts: only one of sixteen parallel-category contrasts survives familywise correction, so a universal ranking benefit is not supported; the remaining intervals do not establish equivalence. Finally, one pinned binary, artifact set, runner, and case set produce byte-identical outputs for all 2,480 pairs on an M4 Max and an A18 Pro host, while mean request latency on the A18 Pro is 4.41–9.33× that on the M4 Max. Supporting controls show that schema exposure changes process RSS and lifecycle changes resident-memory and end-to-end measurements without changing paired outputs. Quality and resource claims therefore require a declared local-agent protocol. We provide the design matrix, minimum reporting record, prospective transfer tests, and independently recomputable evidence.

1 INTRODUCTION

Local SLM agents are commonly evaluated as a model-selection problem: choose a small model, quantize it, and report whether it calls the correct function. That description omits much of the deployed system. A request contains an instruction, a serialized set of tool schemas in a particular order, a response contract, and often a selection rule deciding which schemas fit in context. The request runs on a host inside a server whose process and cache lifetime affect memory and timing. An evaluator then determines what counts as a valid call. Changing any of these components can change quality or measured cost without changing model weights.

The paper’s central result is that output quality can transfer while deployment cost does not. Three findings make that result concrete. First, under one pinned M4 stack, changing the declared request protocol, including its instruction and requested JSON shape, moves exact match by 20.0 points on 0.5B simple and 16.0 points on 0.5B multiple, and reverses the two Qwen artifacts’ ranking when abstention must be decided rather than supplied. Second, a positive schema-order pattern in all twelve earlier simple/multiple contrasts survives familywise correction in only one of sixteen prospectively frozen parallel-category contrasts; the transfer evidence is mixed, and the remaining intervals are not evidence of equivalence. Third, a pinned two-host control produces 2,480/2,480 byte-identical outputs while mean request latency on the A18 Pro is 4.41 to 9.33 times that on the M4 Max. These statements retain explicit protocol scope, but each is positive, reproducible, and more informative than a model-only score.

This paper treats the complete local SLM agent protocol as the system under test. Its methodological contribution is an identification discipline: declare the protocol, vary one component where possible, retain all attempts, freeze prospective schedules, and independently recompute admitted results. Tool-call generation is controlled because expected outputs, schema exposure, and serialization can be manipulated exactly. It is not all of agent behavior: we do not execute tools, assess multi-turn recovery, or claim general agent utility. We ask which conclusions survive when context construction, evaluation, and serving state are explicit.

We organize the evidence around three systems axes:

- **Context construction:** schema exposure, selected content, serialization order, and the instruction/output contract;
- **Serving state:** persistent versus restart-every-20 lifecycle and the resource endpoint used to compare them; and
- **Substrate:** artifact/adaptation contrasts and matched transfer between two Apple host systems.

Our contributions are:

- (1) a protocol-first methodology using label-blind prompts, all-attempt denominators, content-addressed inputs, explicit metric scopes, blocked schedules, independent recomputation, and a minimum reporting checklist;
- (2) a build- and hash-pinned five-category measurement design with explicit protocol, metric, and evidence boundaries;
- (3) a frozen-before-treatment 2,480-case M4 request-protocol control that jointly varies instruction content and requested JSON shape, plus a matched two-host control spanning 2,480 paired cases and two Qwen artifacts;
- (4) schema-exposure, fixed-content order, prospective transfer, artifact, and lifecycle controls that test how far each finding transfers; and
- (5) a public evidence package with compact paired outcomes, frozen protocols, independent recomputation, and a minimum reporting record.

The paper introduces no new model, retriever, training method, or benchmark. Its host-transfer claim is limited to these two Apple hosts, two Qwen artifacts, five category files, and the pinned protocol. Its value is identification discipline: a numerical result is attached to the complete protocol that produced it, and a contrast is named for the factors it actually isolates.

2 RELATED WORK

2.1 Function calling and on-device agents

Gorilla connects language models to large API collections and introduces APIBench [16]. BFCL broadens function-calling evaluation to serial, parallel, and abstention behavior [15]. Hammer studies function masking for robust on-device calling [10]; TinyAgent combines small models, retrieval, and quantization for a MacBook assistant [5]; MobileLLM studies sub-billion on-device models [12]. These systems optimize models or complete agents. We instead isolate measurement-protocol components around fixed local artifacts.

2.2 Schema retrieval and order

Re-Invoke rewrites invocations for zero-shot tool retrieval [4]. ToolRet finds that generic retrieval strength does not imply strong tool retrieval [18]. Less Is More reports that reducing exposed tools can improve edge function-calling efficiency [14]. Our BM25 condition is a measurement probe, not a competitive retriever.

Context order is independently consequential. Lost in the Middle shows that long-context use is position-sensitive [11]; BiasBusters reports earlier-listed-tool preference in controlled tool selection [3]. A top- k experiment that changes both the selected multiset and its order therefore bundles at least two treatments.

2.3 Prompt format and evaluation stability

FormatSpread demonstrates that meaning-preserving prompt-format choices can substantially change model scores and even comparative conclusions [17]. Multi-prompt evaluation similarly shows that single-template benchmark results can be brittle in both absolute and relative terms [13].

Table 1. Protocol-component design matrix for the eleven evidence families. V is varied by the comparison; B is crossed or blocked but fixed within each estimate; F is fixed and recorded; U is fixed but its exact record is unavailable; $-$ is not applicable. $V^\dagger$ marks a varied artifact whose retained byte identity is incomplete. The energy row is an unlicensed audit, so its V records manipulation rather than an identified effect.

Evidence family	H	M	A	I	S	R	O	E	L
Pre-checklist Qwen artifacts	F	$V^\dagger$	U	F	F	$-$	F	F	F
Matched MLX adaptation	F	V	F	F	F	$-$	F	F	F
Same-stack artifacts	F	V	F	F	F	$-$	F	F	F
Ordered energy audit	F	F	F	F	V	F	F	F	F
Balanced route grid	F	F	F	F	V	V	F	F	F
Schema content/order controls	F	F	F	F	V	B	F	F	F
Supporting lifecycle control	F	B	F	F	F	$-$	F	F	V
Prospective order transfer	F	B	F	F	V	B	F	F	F
Prospective lifecycle extension	F	B	F	F	F	$-$	F	F	V
Request-protocol control	F	B	F	V	F	$-$	V	F	F
Matched host transfer	V	B	F	F	F	$-$	F	F	F

Our request-protocol contrast is not a new claim that prompts matter. It tests how that sensitivity appears in local tool use when artifacts, cases, evaluator, API settings, and lifecycle are pinned, and when a template changes the estimand by supplying relevance or call count.

2.4 Serving and resource accounting

PagedAttention addresses KV-cache fragmentation [9]; SGLang optimizes structured language-model programs [20]; LLM in a Flash studies inference under limited memory [1]. We ask a complementary endpoint question: what quality and local resource measurements accompany a complete tool-using SLM protocol?

3 SYSTEM BOUNDARY AND EVIDENCE

3.1 Protocol as treatment

We define

$$P = (H, M, A, I, S, R, O, E, L),$$

where H is the host system, M a model artifact, A its runtime and quantization path, I the instruction, S the serialized schema multiset and order, R an optional selection rule, O the output contract and decoding settings, E the evaluator, and L the server lifecycle. Measured quality is $Q(P, D)$ for fixed cases D . A contrast identifies one component only when all remaining tuple entries are fixed or explicitly blocked. Otherwise we call it an artifact, host-system, or protocol contrast.

This distinction prevents four category errors. Files with the same nominal quantization label need not be identical artifacts. A retrieval ranking changes order as well as content unless the selected multiset is reordered in a control. A persistent process and repeated fresh processes are different serving systems, even when their outputs match. A host contrast can bundle chip, memory, OS, driver, run order, and ambient state even when the executable hash is fixed.

Table 1 applies the tuple to every evidence family. It makes two boundaries visible: the early Qwen comparison predates the minimum record and therefore has incomplete artifact/runtime provenance, while the newer controls fix and record every non-treatment component used by their licensed contrast.

The schema content/order controls keep lifecycle fixed. The ordered energy audit keeps the instruction and BM25 selection procedure fixed while changing the exposed schema width; because

capture order, idle subtraction, and replication are inadequate, that manipulation licenses no energy effect. The request-protocol control varies both *I* and *O* and is therefore not an instruction-only contrast.

3.2 Hosts, runtimes, and artifacts

Most admitted measurements run on an Apple M4 Max (Mac16,5) with 128 GiB unified memory. A matched transfer control repeats a bounded Qwen subset on a MacBook Neo (Mac17,5) with Apple A18 Pro and 8 GiB unified memory. System Profiler reports 40 and 5 GPU cores on these hosts, respectively; this is descriptive context, not an identified explanation for their timing difference. The $8\times$ core-count ratio neither equals nor explains the observed request-latency ratios, which range from 4.41 to 9.33. GGUF artifacts are served through a content-addressed llama-server path [6]. The cross-host control uses the same static build 9870 binary, /usr/bin/python3 3.9.6, and matching binary, runner, model, and case SHA-256 values on both native arm64 macOS hosts. Each admitted execution condition reports AC power at launch; the evidence package retains the full runtime commit.

The matched trial commands leave --n-gpu-layers unset; build 9870 defines that default as auto. The level-3 trial logs do not retain the resolved layer assignment. Retrospective startup-only probes with the same binary and the largest matched artifact select Metal device MTL0 and offload 29/29 layers on both hosts. These probes verify automatic-offload behavior at probe time; they are not trial-time telemetry. Primary pre-checklist artifact evidence uses Qwen2.5 instruction artifacts [19]; same-stack controls include GLM-4 9B Q4_0 and Llama 3.1 8B Q4_K_M [7]. The matched adaptation control uses a pinned Qwen2.5 snapshot under MLX/MLX-LM [2] and two LoRA adapters [8]. Model files, adapters, case files, and the newer executables are identified by SHA-256. Missing exact build provenance for the earliest Qwen rows is retained as a limitation rather than reconstructed.

3.3 Cases, prompts, and evaluation

The workload imports every record, in source order and with zero skips, from five BFCL-v3 single-turn category files: 400 simple, 200 multiple, 200 parallel, 200 parallel-multiple, and 240 irrelevance cases [15]. The converter joins each question file to its possible-answer file by ID (falling back to source position), normalizes tool schemas and accepted argument alternatives, and treats the complete irrelevance file as no-tool-call cases. The resulting files and converter are hash-pinned. These are local protocol measurements, not official BFCL leaderboard scores. The label-blind studies use a category-agnostic instruction that requests a JSON array of zero, one, or multiple tool-call objects and never reveals expected cardinality, relevance, or category. The standard conditioned family instructs the model to choose exactly one or a specified number of calls, or explicitly states irrelevance; for single-call conditions it requests an object rather than the label-blind array. The request-protocol study jointly changes instruction content and requested JSON shape while holding the API response-format setting, decoding parameters, and evaluator fixed. The label-blind protocol jointly measures relevance/cardinality judgment and call generation. The conditioned protocol instead measures instruction compliance and call generation after that decision is supplied; its irrelevance and parallel-category scores are not autonomous abstention or call-count-selection measurements. Temperature is zero. JSON response repair is disabled in the pre-checklist schema grids, prospective studies, and matched host control; invalid outputs or retrieval misses remain in the denominator.

Exact match requires the expected call multiset, function names, argument names, and argument values under the frozen schema-aware evaluator. Parse validity only means a payload can be extracted and validated; it is not strict raw-response JSON compliance.

We use *execution condition* for one fully specified run configuration and *contrast cell* for one reported paired estimand, such as an artifact–category comparison. Schema-grid counts refer to execution conditions; prompt and host tables report contrast cells.

3.4 Minimum reporting checklist

Minimum local-agent protocol record. Every reported result should publish:

- (1) **Artifact** (M, A). Exact revision/file, quantization, adapters, and full SHA-256.
- (2) **Runtime/host** (H, A). Server build or digest, backend and offload policy, hardware, memory, architecture, and OS.
- (3) **Context** (I, S, R). Instruction, schema pool, selection, exposed count, and order.
- (4) **Output** (O). Limits, response contract, decoding, repair, and retries.
- (5) **Evaluation** (E, D). Case/evaluator digests, denominator, metric, unit, and multiplicity.
- (6) **Lifecycle/resources** (L ; **scope in** H). Persistence, cache, batching, endpoint, scope, and sampling.

3.5 Metric scopes and statistical units

On identical case IDs, artifact differences use paired intervals and exact McNemar tests with Holm correction within declared families. In schema studies, five deterministic perturbation seeds share each source query; the inferential unit is therefore the query cluster, not the raw attempt. We average seed-level paired differences within query, use deterministic query-cluster bootstrap intervals, and apply exact cluster sign-flip tests with Holm adjustment. “Deterministic bootstrap” means pseudorandom resampling with a fixed, content-derived seed so an independent checker reproduces the same draws and endpoints byte-for-byte.

Displayed percentages and percentage-point differences use one decimal and round ties to even (for example, 35.25 becomes 35.2); calculations and tests use unrounded counts and statistics.

The request-protocol control pairs exact-match outcomes by case within each artifact–category contrast cell. Label-blind-minus-conditioned differences use deterministic paired case-bootstrap 95% intervals and two-sided exact McNemar tests, with Holm adjustment across the declared ten-cell family. A secondary five-category family compares the two artifacts within the pinned label-blind arm. We keep these as separate families because they test distinct estimands: request-protocol effects within artifact–category contrast cells versus artifact gaps within one prompt arm. The latter remains explicitly secondary and is not used to enlarge the primary template-effect claim. All attempted cases remain in the denominator. Post hoc exact-given-parse-valid rates decompose observed failures descriptively. Parse validity is post-treatment, so conditioning on it can induce selection or collider bias; these rates are not a second causal estimand. Multiplicity families are declared per estimand and study; we do not apply one paper-wide correction across distinct questions. Accordingly, no collection of section-level adjusted results should be read as one omnibus family.

Request latency is client wall time for serving, including prompt evaluation, decoding, local transport, and retries, but excluding launch. End-to-end time includes launch, readiness, requests, settle periods, and shutdown. Peak RSS is the maximum sampled RSS of the launched server parent process. It is neither device-wide unified memory nor whole-system memory, and it does not separate file-backed resident pages from anonymous allocations. Lifecycle systems units are complete condition sessions; repeated task cases within a session are not independent systems replicates.

For the cross-host control, identical case IDs are paired for exact-match and raw-output agreement. Restart-every-20 blocks are the descriptive resampling unit. Deterministic paired bootstrap intervals summarize the A18-Pro/M4-Max ratio of mean request latency and the median paired

Table 2. Claim language is determined by what the comparison identifies.

Evidence	Results	Licensed claim	Prohibited shortcut
Pinned label-blind Qwen artifacts	Sec. 8	The 1.5B artifact is 23.5–45.4 points more exact across five category files under build- and hash-pinned settings.	A causal parameter-count effect or an official BFCL score.
Pre-checklist label-blind Qwen matrix	App. A	The 1.5B artifact is more exact on these category files under the pre-checklist label-blind protocol, with about twice peak process RSS.	A causal parameter-count or uniform “Q4” effect; an official BFCL score.
Pinned M4 request-protocol control	Sec. 6	Changing the declared instruction and requested JSON shape changes non-irrelevance exact match by up to 20.0 points under fixed build, artifact, case, 96-token cap, evaluator, and lifecycle settings; label-blind abstention reverses the artifact ranking.	Attributing the result to instruction alone, exporting it to other prompts, or treating conditioned irrelevance as autonomous abstention.
Same-stack family controls	Sec. 8	Content-addressed artifacts differ substantially under the same prompt and server path.	A causal architecture, training-family, or quantization effect.
Matched MLX control	Sec. 8	These two LoRA recipes do not pass promotion over this matched control.	LoRA cannot improve tool calling.
Base route matrix	Sec. 4	Schema configuration changes quality and process RSS.	BM25 selection alone causes the observed difference.
Fixed-content order controls	Sec. 4	Ranked/sorted order has a positive point estimate in all twelve measured cells; four contrasts survive Holm correction.	Treating five seed perturbations as independent queries or attributing order gains to selection.
k /pool controls	Sec. 4	Canonical top-16 never improves over top-8 here; one decline survives correction. Pool-size effects are mixed.	A pure token-count effect or a universal top-8 optimum.
Prospective order transfer	Sec. 5	One of sixteen contrasts survives familywise correction; the rest remain interval-bounded, not equivalent to zero.	A universal ranking benefit, a null effect, or an equivalence claim.
Lifecycle control	Sec. 7	Exact-match outcomes agree while persistence trades higher peak process RSS for fewer model loads.	Total-memory, general equivalence, or universal latency superiority.
Matched host control	Sec. 9	All 2,480 paired raw outputs agree under the pinned binary and protocol; latency and parent-process RSS retain host-specific ratios.	A pure chip effect, broad hardware equivalence, or whole-device memory comparison.
Ordered energy audit	Sec. 4	No confirmatory energy claim is licensed; the traces are retained only to expose the failed identification boundary.	A width effect, whole-system energy, or cross-device efficiency.

block-level peak-RSS ratio. The raw ratio of observed maximum RSS is reported separately without an interval. These summaries condition on the executed blocks and do not model host order, OS/driver, thermal, or ambient-state uncertainty.

Table 3. Seven immutable baseline evidence families. Counts and status text are regenerated from the hash-validated compact M4-v1 bundle; four controlled extensions described in the text bring the study total to eleven.

Evidence family	Validated completeness
Artifact/category matrix	10/10 category-model cells; 2480/2480 case-model evaluations
Matched MLX control	6/6 matched conditions; 2/2 primary paired adapter comparisons
Same-stack artifact controls	8/8 artifact-category cells; 2400/2400 case-artifact evaluations; 2/2 paired GLM-Llama contrasts
Ordered energy audit (ancillary)	20 traces retained in the ancillary audit; no confirmatory energy claim admitted
Balanced route grid	90/90 balanced route cells measured; 90/90 predeclared plan rows
Schema content/order controls	160/160 control cells; 6400/6400 control attempts; 8800/8800 prompt-token rows; 160/160 predeclared plan rows
Lifecycle control	32/32 sessions; 9600/9600 attempts; 256/256 launches; 7/7 hash-bound analysis artifacts

3.6 Evidence tiers and integrity

The study combines a hash-validated seven-family M4-v1 baseline with four newer extensions. Table 3 records the baseline completeness boundary. The order and lifecycle studies add 6,400 and 12,800 attempts, respectively; their treatment schedules, inputs, claim templates, and expected denominators were prospectively frozen. The request-protocol control adds 2,480 label-blind attempts paired to an existing 2,480-attempt conditioned M4 matrix. Its treatment commands, complete source/input/comparator hashes, denominator, estimands, bootstrap method, and ten-test Holm family were frozen after the comparator was known but before label-blind execution. The matched host control adds 4,960 host attempts (2,480 paired cases). Its commands, input hashes, denominator, and claim boundary were fixed before execution. Its compact audit export and the alignment of the cross-host resampling summaries were finalized after outcomes were available. We therefore use a median paired block-RSS estimand with a descriptive bootstrap interval, report the raw observed maximum separately, and do not call the interval preregistered inference. Generated and independently recomputed artifacts must agree before any extension enters the manuscript.

Here, “historical” means our own pre-freeze baseline/control measurements, not production logs, scraped traces, or third-party measurements.

4 SCHEMA CONFIGURATION EFFECTS

4.1 Exposure changes quality and memory

Every schema experiment in this section uses the same content-addressed Qwen2.5 0.5B Instruct Q4_K_M GGUF artifact. The base route matrix crosses simple and multiple categories, three tool-pool additions, three exposure policies, and five perturbation seeds: 90 measured execution conditions and 3,600 attempts. Full pool exposes every schema; BM25 top-8 and top-16 expose ranked subsets. Gold recall is diagnostic: retrieval misses remain denominator failures.

At multiple/+32, full-pool prompting yields 44/200 exact attempts, versus 118/200 for ranked top-8 and 120/200 for ranked top-16. The pattern is not monotonic: at simple/+128, top-16 yields 87/200 versus 89/200 for full pool (top-8 yields 100/200). At +128 tools, full-pool peak process RSS is 3,451–3,491 MiB; top-8 remains at 961–975 MiB and top-16 at 1,139–1,162 MiB. These descriptive contrasts establish that schema configuration changes the measured system, but they do not isolate retrieval: BM25 changes selected content and order together.

Table 4. Complete base schema-pressure matrix for the content-addressed Qwen2.5 0.5B Q4_K_M GGUF artifact. Each value is exact matches over 200 schema-perturbation attempts (40 unique queries $\times$ five seeds). Parentheses give target-tool retrieval recall; full pool necessarily has 200/200 recall. Values are descriptive because BM25-selected schemas are also BM25-ordered.

Category	Added tools	Full pool	BM25 top-8	BM25 top-16
simple	0	155/200	155/200 (200/200)	155/200 (200/200)
	32	78/200	125/200 (200/200)	124/200 (200/200)
	128	89/200	100/200 (197/200)	87/200 (200/200)
multiple	0	115/200	125/200 (200/200)	125/200 (200/200)
	32	44/200	118/200 (198/200)	120/200 (200/200)
	128	52/200	101/200 (195/200)	101/200 (195/200)
<i>Peak RSS at +128 tools (MiB; min-max over five cells):</i>				
simple	128	3450.5–3461.4	960.9–970.2	1138.5–1151.7
multiple	128	3483.4–3491.4	971.3–975.1	1153.6–1161.9

Schema load and process memory

Control grid ranges across category/policy conditions - separate zero-based panels

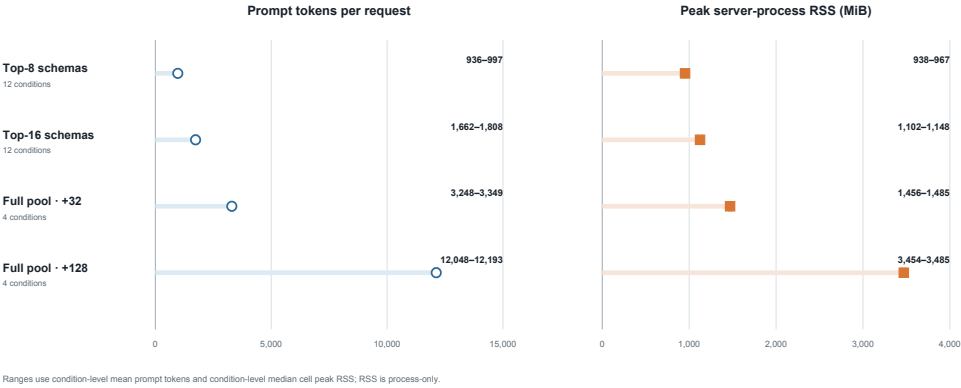


Fig. 1. Schema exposure changes prompt tokens and sampled parent-process RSS. Both panels use the Qwen2.5 0.5B Q4_K_M GGUF artifact and start at zero. These ranges come from the 160-cell fixed-content control grid; Table 4 reports the separate 90-cell base grid.

4.2 Historical fixed-content order controls

The historical control grid contains 160 execution conditions and 6,400 attempts over 40 queries per category, five perturbation seeds, +32/+128 pools, and eight policies. Companion rows bring prompt-token accounting to 8,800 rows. Every reported order pair has an identical schema multiset and identical mean prompt tokens; only serialization changes.

Ranked or sorted order has a positive point estimate in all twelve deployable simple/multiple contrasts, from +7.5 to +24.5 points. Four survive Holm correction: multiple/+128 full pool (+24.5), multiple/+32 top-16 (+20.5), simple/+32 top-16 (+19.0), and simple/+32 full pool (+17.0). Several additional intervals exclude zero without surviving the twelve-test family. The direction is therefore broadly consistent in this historical grid, while familywise evidence is concentrated in four cells.

Table 5. Deployable fixed-content order contrasts for the same content-addressed Qwen2.5 0.5B Q4_K_M GGUF artifact. Rates are over all 200 attempts (40 source-query clusters $\times$ five blocked schema perturbations). Differences use query-cluster bootstrap 95% intervals; p -values are exact query-cluster sign-flip tests with Holm correction over the twelve-row family. Every pair has an identical schema multiset and identical mean prompt tokens.

Category	Pool	First order – second order	First	Second	Difference (pp; 95%)	Holm p
multiple	+128	top-16 ranked – canonical	50.5	32.0	+18.5 [+5.5, +32.0]	0.0927
multiple	+128	top-8 ranked – canonical	50.5	43.0	+7.5 [–3.5, +19.0]	0.6962
multiple	+128	full-pool sorted – canonical	38.5	14.0	+24.5 [+10.0, +39.5]	0.0235
multiple	+32	top-16 ranked – canonical	60.0	39.5	+20.5 [+10.0, +31.0]	0.0061
multiple	+32	top-8 ranked – canonical	59.0	46.0	+13.0 [+4.5, +22.5]	0.0535
multiple	+32	full-pool sorted – canonical	26.5	14.0	+12.5 [+3.5, +22.5]	0.1142
simple	+128	top-16 ranked – canonical	43.5	34.5	+9.0 [–7.0, +25.0]	0.6962
simple	+128	top-8 ranked – canonical	50.0	40.5	+9.5 [–5.0, +24.0]	0.6962
simple	+128	full-pool sorted – canonical	39.5	22.5	+17.0 [+1.5, +33.0]	0.2486
simple	+32	top-16 ranked – canonical	62.0	43.0	+19.0 [+10.5, +28.5]	0.0019
simple	+32	top-8 ranked – canonical	62.5	52.0	+10.5 [+1.0, +20.5]	0.2486
simple	+32	full-pool sorted – canonical	31.0	14.0	+17.0 [+9.5, +25.5]	0.0095

Schema order effects at fixed schema content

Ranked/BM25 order minus canonical · 95% query-cluster bootstrap intervals · 40 clusters · 5 blocked seeds

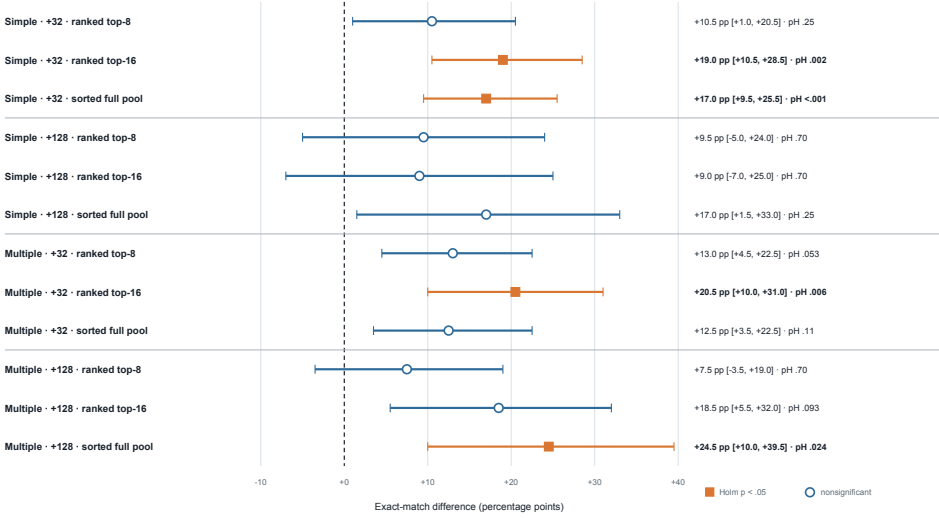


Fig. 2. Historical fixed-content order effects on simple and multiple cases. All cells use the Qwen2.5 0.5B Q4_K_M GGUF artifact. Intervals are query-cluster bootstrap 95% intervals; orange squares survive Holm correction within the twelve-contrast family.

4.3 Width and distractor-pool controls

The four paired rows in the ancillary file `route_v2_control_comparisons.csv` show that canonical top-16 adds 777.0, 789.1, 795.3, and 810.5 mean prompt tokens over canonical top-8. With order fixed, top-16 improves none of four measured cells; differences range from –6.0 to –11.0 points. The simple/+32 decline of 9.0 points survives four-test Holm correction. These are width-and-content

Table 6. Deployable k and distractor-pool contrasts under canonical order for the same content-addressed Qwen2.5 0.5B Q4_K_M GGUF artifact. The first four rows compare top-16 minus top-8 (four-test Holm family); the remaining rows compare +128 minus +32 tools (eight-test Holm family). Query-cluster bootstrap 95% intervals and exact sign-flip tests use 40 source-query clusters with five seeds blocked within query.

Factor	Category	Pool/policy	First	Second	Difference (pp; 95%)	Holm p
k	multiple	+128	32.0	43.0	-11.0 [-22.0, 0.0]	0.2186
k	multiple	+32	39.5	46.0	-6.5 [-14.5, +1.0]	0.2788
k	simple	+128	34.5	40.5	-6.0 [-18.5, +5.5]	0.3809
k	simple	+32	43.0	52.0	-9.0 [-15.5, -3.0]	0.0438
Pool	multiple	top-16 canonical	32.0	39.5	-7.5 [-15.5, +0.5]	0.5801
Pool	multiple	top-8 canonical	43.0	46.0	-3.0 [-10.0, +3.0]	0.9409
Pool	multiple	full-pool sorted	38.5	26.5	+12.0 [0.0, +24.5]	0.5646
Pool	multiple	full-pool canonical	14.0	14.0	0.0 [-10.0, +10.0]	1.0000
Pool	simple	top-16 canonical	34.5	43.0	-8.5 [-19.0, +2.0]	0.5801
Pool	simple	top-8 canonical	40.5	52.0	-11.5 [-23.0, 0.0]	0.5646
Pool	simple	full-pool sorted	39.5	31.0	+8.5 [-1.0, +18.5]	0.5801
Pool	simple	full-pool canonical	22.5	14.0	+8.5 [-4.0, +22.5]	0.7555

effects, not pure token-count effects, because the additional schemas also change candidate content. Expanding the distractor universe from +32 to +128 produces mixed effects; none survives its eight-test family.

5 PROSPECTIVE SCHEMA-ORDER TRANSFER

5.1 Design

The prospective transfer test applies the fixed-content order treatment to two Qwen artifacts, parallel and parallel-multiple categories, +32/+128 pools, top-8/top-16 widths, and five blocked schema seeds. Forty source queries per category were selected by a frozen SHA-256 ranking before treatment execution. Every order pair preserves both the selected multiset and prompt-token count. The alternating ranked-first/canonical-first schedule contains 160 cache-isolated execution conditions and all 6,400 planned attempts.

Historical and prospective schema studies both disable JSON response repair. The between-study comparison nevertheless changes task category and study epoch and adds a second artifact; it tests transfer but does not factorially identify why the historical pattern weakens.

For each of sixteen declared contrasts, the endpoint is ranked minus canonical exact-match rate. Five seed-level paired differences are aggregated within each query. We report deterministic query-cluster bootstrap intervals and exact two-sided cluster sign-flip tests, with Holm correction across the full family.

5.2 Mixed evidence under limited resolution

The result does not support a uniform positive order claim. For the 0.5B artifact, all eight point estimates are positive (+4.0 to +14.0 points), and the parallel/+32/top-16 contrast survives correction (+14.0 points, Holm-adjusted $p = 0.0374$). For the 1.5B artifact, parallel estimates are nonnegative, but parallel-multiple estimates range from -5.0 to +0.5 points. Only one of sixteen contrasts survives familywise correction.

Failure to reject the other fifteen contrasts is not evidence of no effect. Their 95% intervals retain positive upper bounds from +4.0 to +17.5 points, and the 0.5B canonical rates are as low as 3.0–21.0%, which compresses the observable negative range. No equivalence margin or prospective power target was declared. We therefore use the reported cluster intervals as the resolution audit rather than adding a post hoc observed-power calculation: the study can reject a uniformly

Table 7. Prospective parallel-category schema-order transfer test. Ranked and Canonical are exact-match percentages over 200 attempts per order (40 query clusters $\times$ five seeds); differences and 95% intervals are percentage points. k is exposed schema width. Selected schema multisets and prompt-token counts are identical within pairs. Non-significant intervals are not equivalence claims.

Artifact	Category	Pool	k	Ranked	Canonical	Difference [95% CI]; Holm p
0.5B Q4	parallel	+32	8	31.0	21.0	+10.0 [+4.0, +17.5]; 0.0586
0.5B Q4	parallel-multiple	+32	8	16.5	12.5	+4.0 [-2.0, +10.5]; 1
0.5B Q4	parallel	+32	16	21.5	7.5	+14.0 [+6.0, +23.0]; 0.0374
0.5B Q4	parallel-multiple	+32	16	10.5	4.0	+6.5 [+2.0, +13.0]; 0.219
0.5B Q4	parallel	+128	8	24.5	17.5	+7.0 [+1.0, +13.5]; 0.677
0.5B Q4	parallel-multiple	+128	8	13.0	7.5	+5.5 [+1.0, +11.0]; 0.727
0.5B Q4	parallel	+128	16	14.0	7.0	+7.0 [0.0, +14.5]; 0.729
0.5B Q4	parallel-multiple	+128	16	11.0	3.0	+8.0 [+1.5, +16.5]; 0.727
1.5B Q4	parallel	+32	8	51.0	44.0	+7.0 [+1.0, +14.0]; 0.727
1.5B Q4	parallel-multiple	+32	8	42.5	42.0	+0.5 [-9.0, +9.5]; 1
1.5B Q4	parallel	+32	16	41.0	41.0	0.0 [-7.5, +8.0]; 1
1.5B Q4	parallel-multiple	+32	16	38.0	42.0	-4.0 [-13.0, +5.0]; 1
1.5B Q4	parallel	+128	8	45.0	38.5	+6.5 [+0.5, +13.5]; 0.729
1.5B Q4	parallel-multiple	+128	8	30.5	32.5	-2.0 [-8.0, +4.0]; 1
1.5B Q4	parallel	+128	16	43.0	37.0	+6.0 [+0.5, +12.0]; 0.729
1.5B Q4	parallel-multiple	+128	16	28.5	33.5	-5.0 [-15.5, +4.0]; 1

transferable benefit, but it cannot establish that most individual effects are zero or smaller than a policy-relevant margin.

The prospective transfer evidence is mixed across declared contrasts. It does not support a uniform benefit of ranked order, while the non-significant intervals are not evidence of zero effect or equivalence.

This transfer test narrows the scientific conclusion. The historical simple/multiple grid shows a consistently positive pattern, but the prospective parallel-category evidence is mixed and underpowered for equivalence. Schema order remains a real protocol variable; a universal benefit of ranking is not supported, while smaller configuration-specific effects remain compatible with the intervals.

6 REQUEST-PROTOCOL EFFECTS

6.1 Frozen treatment and paired analysis

The existing M4 host-control matrix supplies the conditioned arm. Before executing the missing label-blind arm, we froze a ten-contrast-cell schedule, all runner sources and commands, binary, model, and case hashes, the complete conditioned comparator snapshot, all-attempt denominators, paired bootstrap intervals, and one ten-test Holm family. A preflight rebuilt every stored conditioned prompt and verified that the treatment preserves each visible schema serialization and user request. Both arms use build 9870, the same Qwen files and SHA-256 values, the same Python 3.9.6 runtime, batch size 20, restart-every-20, temperature zero, 96 output tokens, no response-format schema, no JSON repair, and the same schema-aware evaluator. The declared treatment is the complete request protocol: conditioned standard versus label_blind_json. It jointly varies instruction content (I) and the requested object-versus-array JSON shape (O), so it is a protocol contrast rather than an identified instruction-only effect.

Prospective schema-order effects

Ranked minus canonical exact match · query-cluster bootstrap 95% intervals · 40 queries per contrast

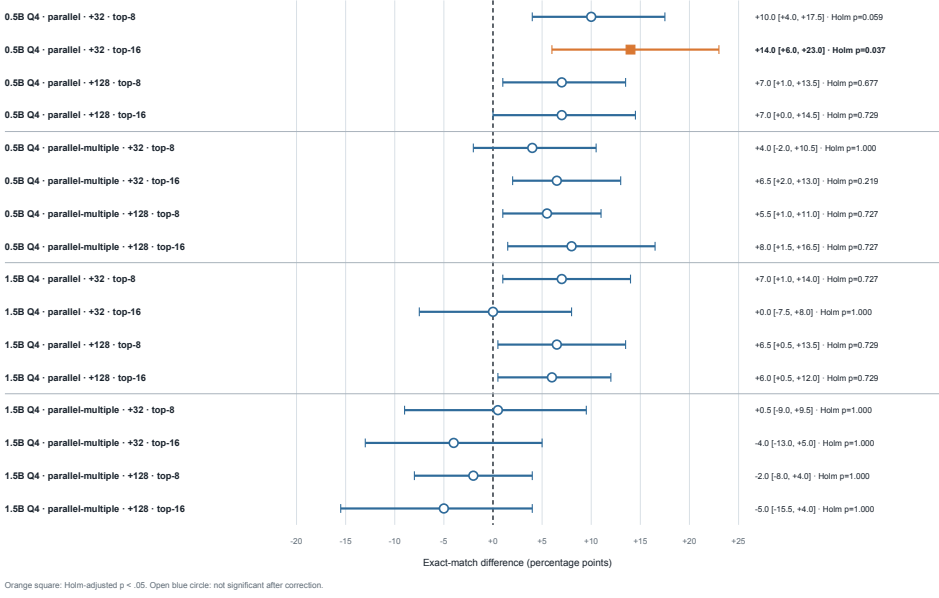


Fig. 3. Prospective fixed-content order effects on parallel categories. Points show ranked-minus-canonical exact-match differences; intervals are query-cluster bootstrap 95% intervals.

Table 8. Frozen-before-treatment M4 request-protocol contrast under build 9870; the treatment jointly changes instruction content and requested JSON shape; the conditioned comparator pre-existed the freeze. Label-blind minus cardinality-conditioned differences use paired case-bootstrap 95% intervals; p_H is the two-sided exact McNemar value adjusted by Holm across all ten artifact–category cells. Artifact, runtime, schema serialization, API response-format, decoding, evaluator, and restart-every-20 settings are fixed. Irrelevance rows are manipulation checks because only the label-blind arm requires autonomous abstention.

Artifact	Category	Attempts	Conditioned exact	Label-blind exact	Difference (pp; 95%; p_H)
0.5B Q4	simple	400	268	188	-20.0 [-25.2, -15.0]; 2.23×10^{-12}
1.5B Q4	simple	400	332	329	-0.8 [-3.0, +1.5]; 1.000
0.5B Q4	multiple	200	130	98	-16.0 [-23.0, -9.5]; 6.34×10^{-5}
1.5B Q4	multiple	200	155	150	-2.5 [-7.0, +1.5]; 1.000
0.5B Q4	parallel	200	42	30	-6.0 [-12.0, +0.0]; 0.403
1.5B Q4	parallel	200	104	83	-10.5 [-17.0, -4.0]; 0.019
0.5B Q4	parallel-multiple	200	19	25	+3.0 [-1.5, +7.5]; 1.000
1.5B Q4	parallel-multiple	200	77	72	-2.5 [-6.0, +1.0]; 1.000
0.5B Q4	irrelevance	240	240	36	-85.0 [-89.6, -80.4]; 7.78×10^{-61}
1.5B Q4	irrelevance	240	211	145	-27.5 [-33.3, -22.1]; 2.44×10^{-19}

The conditioned arm ran first and was known when the treatment protocol was frozen; the label-blind arm ran later the same day in the same model/category order. Thus the treatment and analysis were frozen before their outcomes, but the two-arm schedule is neither randomized nor counterbalanced. This supports a bounded request-protocol contrast for exact match. Table 9 retains latency and RSS diagnostics, but fixed arm order and different response lengths prevent a causal cost interpretation.

Table 9. Post hoc format, output-limit, and latency diagnostics for the same pinned M4 request-protocol contrast cells. Parse-valid and cap entries report conditioned/label-blind counts with percentages. Exact/valid reports exact matches over schema-valid attempts in each arm. Latency is label-blind divided by conditioned mean request latency. Conditional rates and latency ratios are descriptive: parse validity is post-treatment, arm order is fixed, and response lengths differ. Reached 96 tokens means the aligned llama-server record decoded exactly the fixed ceiling; it is an operational cap-reach diagnostic, not a recovered API finish_reason.

Artifact	Category	Parse valid C/LB	Exact/valid C	Exact/valid LB	96-token cap C/LB	Latency LB/C
0.5B Q4	simple	392/286 (98.0/71.5%)	268/392 (68.4%)	188/286 (65.7%)	2/3 (0.5/0.8%)	1.188
1.5B Q4	simple	400/394 (100.0/98.5%)	332/400 (83.0%)	329/394 (83.5%)	0/0 (0.0/0.0%)	1.352
0.5B Q4	multiple	199/161 (99.5/80.5%)	130/199 (65.3%)	98/161 (60.9%)	1/1 (0.5/0.5%)	1.658
1.5B Q4	multiple	199/196 (99.5/98.0%)	155/199 (77.9%)	150/196 (76.5%)	0/0 (0.0/0.0%)	1.531
0.5B Q4	parallel	155/115 (77.5/57.5%)	42/155 (27.1%)	30/115 (26.1%)	54/51 (27.0/25.5%)	0.983
1.5B Q4	parallel	122/132 (61.0/66.0%)	104/122 (85.2%)	83/132 (62.9%)	75/60 (37.5/30.0%)	1.091
0.5B Q4	parallel-multiple	155/141 (77.5/70.5%)	19/155 (12.3%)	25/141 (17.7%)	46/39 (23.0/19.5%)	1.143
1.5B Q4	parallel-multiple	120/135 (60.0/67.5%)	77/120 (64.2%)	72/135 (53.3%)	82/66 (41.0/33.0%)	1.218
0.5B Q4	irrelevance	240/200 (100.0/83.3%)	240/240 (100.0%)	36/200 (18.0%)	0/3 (0.0/1.2%)	4.670
1.5B Q4	irrelevance	211/237 (87.9/98.8%)	211/211 (100.0%)	145/237 (61.2%)	0/1 (0.0/0.4%)	2.787

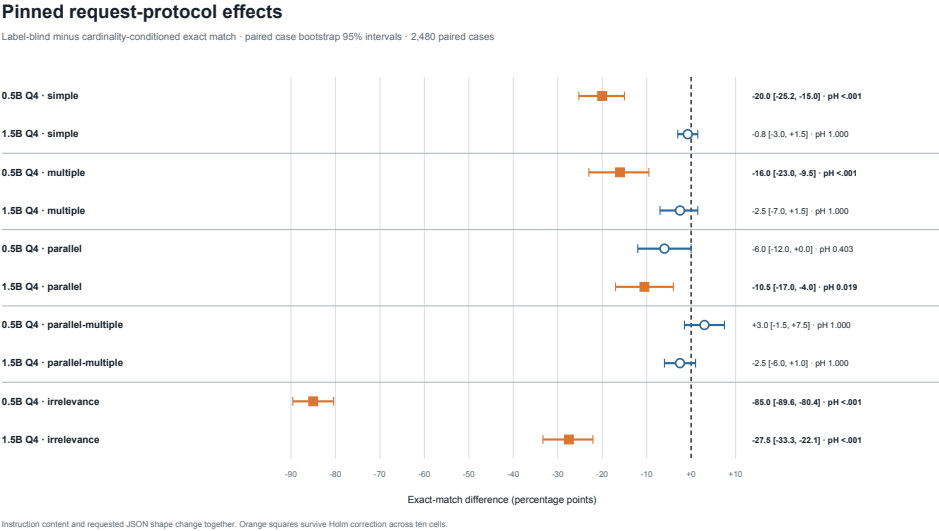


Fig. 4. Pinned label-blind minus cardinality-conditioned exact-match effects. Bars are deterministic paired case-bootstrap 95% intervals. Orange squares survive Holm correction across the ten artifact–category contrast cells. The contrast changes the complete request protocol, including cardinality disclosure and requested JSON shape, rather than one isolated sentence.

6.2 Request-protocol effects and an abstention ranking reversal

On the four category types where both prompts still require a tool-call decision, three effects survive the declared ten-cell Holm correction: 0.5B simple (−20.0 points, 95% CI [−25.2, −15.0]), 0.5B multiple (−16.0, [−23.0, −9.5]), and 1.5B parallel (−10.5, [−17.0, −4.0]). The remaining non-irrelevance effects range from −6.0 to +3.0 points and do not survive correction. Both irrelevance contrasts also survive correction, but their raw magnitudes are reported as manipulation checks

because the conditioned instruction supplies irrelevance while the label-blind instruction requires an autonomous abstention decision.

On the simple and multiple call-generation files, request-protocol sensitivity is concentrated in the smaller artifact: the 0.5B effects are -20.0 points on simple and -16.0 on multiple, while the corresponding 1.5B effects are -0.8 and -2.5 points with intervals spanning zero. Within these two pinned artifacts and category files, the smaller artifact is more protocol-fragile. This is a descriptive two-artifact pattern, not a general parameter-size effect or a predeclared artifact-by-protocol interaction.

Table 9 provides a post hoc audit of two failure channels. Schema validity moves in both directions: for example, 0.5B parallel falls from 77.5% conditioned to 57.5% label-blind, whereas 1.5B parallel-multiple rises from 60.0% to 67.5%. Reaching the 96-token ceiling occurs in 23.0–41.0% of conditioned and 19.5–33.0% of label-blind parallel/parallel-multiple attempts. The conditioned arm reaches it more often in all four cells, so extra label-blind cap reaches cannot alone explain the drops; nevertheless, these effects include request-protocol-output-limit interaction and are not truncation-free judgment effects.

Exact among parse-valid attempts separates two descriptive channels. For 0.5B simple and multiple, raw differences of -20.0 and -16.0 points narrow to -2.6 and -4.5 points after restricting each arm to its valid payloads; the large raw drops accompany much larger request-protocol-associated parse-validity gaps. For 1.5B parallel, validity rises from 61.0% to 66.0% while exact among valid payloads falls from 85.2% to 62.9%, a -22.4 -point difference, so format failure cannot explain that cell. On irrelevance, the conditional differences remain -82.0 points for 0.5B and -38.8 for 1.5B. These conditional rates are post-treatment selections that can incur collider bias, not causal effects of format compliance; the multi-call rates also remain inside the disclosed 96-token interaction.

The irrelevance manipulation check exposes the estimand difference directly. Conditioned parse-valid counts equal exact counts for both artifacts: 240/240 for 0.5B and 211/240 for 1.5B. Thus every conditioned payload that parses is correct, and the 1.5B artifact's 29 misses are all schema-invalid outputs rather than wrong calls among valid payloads. Label-blind 36/240 and 145/240 require the model to decide abstention. Under that label-blind arm, 1.5B exceeds 0.5B by 23.5–45.4 points in all five category files; all five secondary artifact contrasts survive their separately declared Holm family. The scientifically useful irrelevance result is therefore the ranking reversal: the conditioned -12.1 -point 1.5B-minus-0.5B gap becomes $+45.4$ points when autonomous judgment is required, not the tautological -85.0 -point within-0.5B request-protocol contrast.

The two label-blind epochs agree on direction but not magnitude. In category order (simple, multiple, parallel, parallel-multiple, irrelevance), 0.5B exact counts are 269, 122, 63, 59, and 31 in the pre-checklist epoch versus 188, 98, 30, 25, and 36 when pinned. The corresponding 1.5B counts are 339, 153, 109, 98, and 119 versus 329, 150, 83, 72, and 145. Accordingly, the artifact gaps widen from 15.5–36.7 points in the pre-checklist epoch to 23.5–45.4 points when pinned. This is not a failed replication: the pre-checklist epoch uses a 256-token limit and an API response-format schema, whereas the pinned epoch uses 96 tokens and no response schema, and the pre-checklist server build and retained artifact hashes are unavailable. The common direction is the portable observation; the magnitude is protocol-specific, and the bundled epoch changes do not identify a cap mechanism.

This does not identify a universal label-leakage penalty. The treatment changes a complete request protocol: disclosure, decision wording, and, for single-call cases, array-versus-object output guidance move together. Schema validity and output length are therefore diagnostics, not separate randomized treatments. The licensed claim is narrower: under this pinned stack, the declared request protocol changes what exact match measures and changes observed quality enough to alter the artifact ranking on irrelevance.

Table 10. Counterbalanced serving-lifecycle control. Ratios are persistent divided by restart-every-20, summarized as median [min, max] over four paired condition sessions per model–category cell. Exact-rate differences are persistent minus restart-every-20. RSS samples the launched server parent process, not total memory.

Artifact	Category	Exact-rate difference	Peak RSS ratio	Request-latency ratio	End-to-end ratio
Qwen2.5 0.5B Q4	multiple	0.000 [0, 0]	2.518 [2.506, 2.532]	1.214 [1.180, 1.224]	0.747 [0.712, 0.753]
Qwen2.5 0.5B Q4	simple	0.000 [0, 0]	3.483 [3.471, 3.503]	1.168 [1.043, 1.462]	0.671 [0.569, 0.874]
Qwen2.5 1.5B Q4	multiple	0.000 [0, 0]	2.622 [2.621, 2.627]	1.135 [0.887, 1.168]	0.823 [0.689, 0.861]
Qwen2.5 1.5B Q4	simple	0.000 [0, 0]	3.561 [3.551, 3.565]	1.142 [0.964, 1.760]	0.802 [0.713, 1.073]

Serving lifecycle: peak process-RSS ratios

Persistent / restart-every-20 · median and min-max across 4 paired blocks · every exact-rate delta = 0

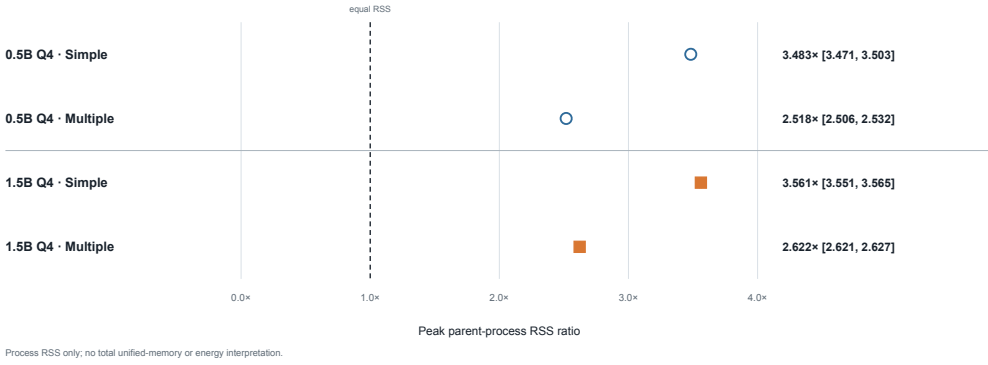


Fig. 5. Supporting persistent/restart-every-20 peak parent-process RSS ratios over four paired blocks per model–category cell.

7 SERVING LIFECYCLE EFFECTS

7.1 Supporting simple/multiple control

The supporting lifecycle study crosses two Qwen artifacts, simple and multiple categories, persistent serving, restart-every-20 serving, and four paired blocks. Its 32 condition sessions retain all 9,600 attempts and 256 launches, with two persistent-first and two restart-first pairs per contrast cell.

At temperature zero, unchanged outputs are expected when lifecycle is the only intended treatment. All 4,800 case pairs agree, verifying that retained serving state, batching, and implementation behavior did not perturb outputs here. Persistent serving reaches 2.52–3.56× peak parent-process RSS, while median end-to-end ratios are 0.67–0.82 because restart-every-20 repeatedly pays launch and readiness costs. Serving-phase request latency moves in the opposite direction and is reported descriptively. Because RSS includes resident file-backed mappings, this endpoint can increase when a persistent process retains more mapped model pages even if allocation demand is unchanged. The runs do not retain anonymous/file-backed or proportional-footprint decomposition.

7.2 Prospective parallel-category extension

The prospective study repeats the lifecycle treatment on parallel and parallel-multiple categories with eight blocked, counterbalanced pairs per contrast cell. It contains 64 condition sessions, 12,800

Table 11. Prospective serving-lifecycle extension on parallel categories. Ratios are persistent divided by restart-every-20 and reported as median [minimum, maximum] over eight paired blocks.

Artifact	Category	Exact difference	Peak RSS ratio	Request-latency ratio	End-to-end ratio
0.5B Q4	parallel	0.000 [0, 0]	2.059 [2.047,2.067]	1.068 [1.025,1.117]	0.829 [0.786,0.869]
0.5B Q4	parallel-multiple	0.000 [0, 0]	2.412 [2.402,2.433]	1.123 [1.094,1.178]	0.858 [0.836,0.872]
1.5B Q4	parallel	0.000 [0, 0]	2.131 [2.120,2.135]	1.051 [1.008,1.133]	0.915 [0.869,0.956]
1.5B Q4	parallel-multiple	0.000 [0, 0]	2.477 [2.474,2.483]	1.095 [1.075,1.108]	0.940 [0.907,0.952]

Prospective serving-lifecycle comparisons

Persistent divided by restart-every-20 - median and min-max across 8 paired blocks - exact-match difference = 0 in every cell

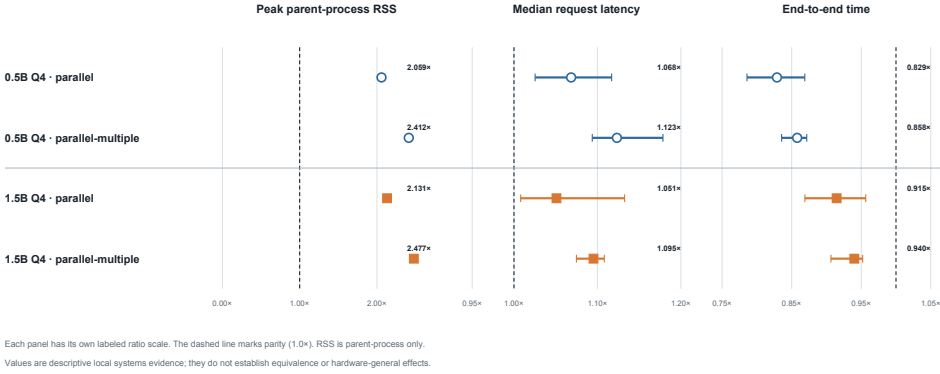


Fig. 6. Prospective lifecycle extension. Ratios compare persistent serving with restart-every-20 over eight paired blocks; exact-match differences are zero in every cell. Each panel uses its own labeled ratio scale.

attempts, and 352 launches. Cases, case order, prompt, decoding, runtime, and evaluator are identical within each condition pair.

The prospective extension repeats that verification: every exact-match difference is zero. Persistent serving uses 2.059–2.477 $\times$ peak parent-process RSS and 1.051–1.123 $\times$ median request latency, while end-to-end ratios are 0.829–0.940. Thus all three systems directions recur: persistence raises process RSS, modestly raises within-serving latency, and lowers total elapsed time. Magnitudes vary across categories and designs, so neither study licenses a universal multiplier. The mechanism is not identified; plausible contributors include retained cache/allocator state, larger resident memory, or different within-session thermal trajectories created by restart gaps. Clock frequency and thermal pressure and RSS mapping classes were not recorded, so the latency and resident memory ratios are descriptive and cannot distinguish these hypotheses.

8 ARTIFACT AND ADAPTATION EFFECTS

8.1 Pinned Qwen artifact contrast

Under build 9870, full artifact, case, and runner hashes, restart-every-20 serving, and the same evaluator, Table 12 places the artifact gaps under both declared request protocols side by side. In category order (simple, multiple, parallel, parallel-multiple, irrelevance), the conditioned 1.5B-minus-0.5B gaps are +16.0, +12.5, +31.0, +29.0, and –12.1 points; the label-blind gaps are +35.2, +26.0, +26.5, +23.5, and +45.4. Thus the request protocol widens the artifact gap by 19.2 and 13.5 points on simple and multiple, narrows it by 4.5 and 5.5 points on the parallel files, and reverses

Table 12. Build- and hash-pinned Qwen artifact gaps under two declared request protocols on the M4 Max. Exact entries are 0.5B/1.5B counts and gaps are 1.5B minus 0.5B. The counts reuse Table 8 observations and are not independent evidence. Label-blind intervals use paired case bootstrap; all five label-blind gaps survive their separately declared Holm family. Conditioned gaps and gap shifts have no intervals because no conditioned artifact-gap or interaction family was declared; gap shifts are post hoc descriptive comparisons.

Category	Attempts	Conditioned		Label-blind		Gap shift (pp)
		Exact	Gap (pp)	Exact	Gap (pp; 95%)	
simple	400	268/332	+16.0	188/329	+35.2 [30.0, 40.5]	+19.2
multiple	200	130/155	+12.5	98/150	+26.0 [17.5, 34.5]	+13.5
parallel	200	42/104	+31.0	30/83	+26.5 [19.0, 34.0]	-4.5
parallel-multiple	200	19/77	+29.0	25/72	+23.5 [17.0, 30.0]	-5.5
irrelevance	240	240/211	-12.1	36/145	+45.4 [38.8, 52.1]	+57.5

Table 13. Label-blind artifact controls on the local llama-server path. GLM/Llama use pinned build 9870; the earlier Qwen rows have matching protocol but no recorded exact build. Latency is omitted because each cell has one sequential execution. The primary paired contrast is GLM minus Llama; it is an artifact contrast, not a causal family effect.

Category	Artifact	Exact	Peak process RSS (MiB)	GLM-Llama difference (pp)
simple	GLM-4 9B Q4_0	347/400	5667.0	+36.5 [31.3, 41.7]
	Llama 3.1 8B Q4_K_M	201/400	6097.6	
	Qwen2.5 0.5B Q4_K_M	269/400	637.4	
	Qwen2.5 1.5B Q4_K_M	339/400	1335.0	
multiple	GLM-4 9B Q4_0	172/200	5830.1	+33.5 [26.3, 40.7]
	Llama 3.1 8B Q4_K_M	105/200	6447.4	
	Qwen2.5 0.5B Q4_K_M	122/200	689.7	
	Qwen2.5 1.5B Q4_K_M	153/200	1423.5	

its sign on irrelevance. This cross-protocol comparison is post hoc and descriptive; no interaction test was declared. Table 12 re-expresses the observations already reported in Table 8; it is not independent evidence.

Under the category-agnostic label-blind protocol, the 1.5B artifact exceeds the 0.5B artifact by 23.5–45.4 points on all five category files. Every paired label-blind artifact contrast survives Holm correction within its separately declared five-category family. The archived BFCL-v3 Table 1, accessed July 30, 2026, reports Qwen2.5 Instruct (Prompt) overall scores of 35.3 for 1.5B and 21.0 for 0.5B [15]; those values are orientation only, not a validation target, because the official handler, artifacts, prompt, output processing, and aggregation differ from this local protocol.

This is an artifact contrast, not a parameter-count ablation: the files differ in size and tensor-level quantization composition. Appendix A retains the earlier pre-checklist matrix. Its common direction is useful, but its 15.5–36.7-point magnitude and RSS ratios are not promoted because exact build and artifact identities were not retained.

8.2 Same-stack artifact controls

On the same local server path, GLM exceeds Llama by 36.5 points on simple and 33.5 points on multiple despite similar 5.5–6.3 GiB process-RSS footprints. The GLM-Qwen 1.5B simple contrast is not significant (+2.0 points, 95% interval [-1.7, +5.7]), while GLM is +9.5 points on multiple

Table 14. Snapshot-pinned MLX simple-category control under the label-blind protocol. Differences are adapter minus matched unadapted control; Holm correction is within the two-adapter primary block.

Condition	Exact	Difference (pp)	Holm p
Unadapted	251/400	–	–
LoRA, LR 5×10^{-6}	232/400	–4.8 [–8.7, –0.8]	0.0248
LoRA, LR 10^{-5}	228/400	–5.8 [–10.0, –1.5]	0.0212

([+4.2, +14.8]; Holm-adjusted $p = 0.0026$ across eight family-versus-Qwen comparisons). These are content-addressed artifact contrasts, not causal effects of architecture, family, size, or quantization.

8.3 Matched LoRA negative control

The pinned MLX study compares two audited 2,000-step Glaive LoRA recipes with their matched unadapted model. The control reaches 251/400 exact on the simple category file, versus 232/400 and 228/400 for learning rates 5×10^{-6} and 10^{-5} . Differences of –4.8 and –5.8 points remain significant after within-protocol Holm correction. This rejects promotion of these recipes under this protocol; it does not imply that LoRA is generally ineffective.

9 MATCHED CROSS-HOST TRANSFER

9.1 Design and admission

We repeat the two Qwen Q4 artifacts on all five category files under one pinned llama-server binary and the same runner, Python version, model files, case files, cardinality-conditioned prompt, evaluator, temperature-zero decoding, and restart-every-20 lifecycle. The M4 Max host runs macOS 26.2 (build 25C56); the MacBook Neo with A18 Pro runs macOS 26.4 (build 25E246). Ten contrast cells per host produce 2,480 attempts per host, or 2,480 paired cases and 4,960 host attempts. The importer admits a cell only after row counts, summaries, batch manifests, provenance, hashes, and AC-at-launch status cross-foot. A compact independent checker then recomputes paired outcomes and all restart-block ratios.

The exact-match columns in Table 15 inherit the cardinality-conditioned prompt. They are valid for the paired byte-identity and resource comparison because both hosts receive the same instruction, but they are not measurements of autonomous relevance, abstention, or call-count selection and should not be compared directly with the label-blind artifact or schema results. The parallel and parallel-multiple counts also inherit the fixed 96-token output ceiling: 23.0–41.0% of the corresponding conditioned M4 attempts reach it in Table 9. Those exact counts are therefore output-budget-bounded observations, not unconstrained capability floors.

This is a host-system transfer contrast, not a pure chip comparison. The hosts differ in chip, memory capacity, OS/driver version, and unmeasured ambient state; the complete M4 matrix also precedes the complete A18 Pro matrix rather than being interleaved. We therefore use the output pairs as a deterministic transfer check and treat latency and RSS ratios as descriptive local systems measurements.

9.2 Outputs transfer; timing does not

All 2,480 paired raw outputs are byte-identical, yielding 100% raw-output agreement and zero exact-match disagreements in every one of the ten contrast cells. Thus, under this pinned binary and deterministic protocol, generated text reproduces exactly across the two hosts even though both retrospective probes select Metal and fully offload the largest artifact. The M4 exact counts are the conditioned column of Table 8 in all ten contrast cells and therefore are not independent

Table 15. Matched cross-host transfer under one pinned binary, identical artifact/case/runner hashes, temperature zero, and restart-every-20 serving. Pairs/blocks reports paired cases and executed 20-case restart blocks. Exact columns are M4 Max/A18 Pro counts under the cardinality-conditioned prompt; they verify paired host agreement and are not label-blind judgment scores. Ratios are computed within each artifact–category cell as A18 Pro divided by M4 Max. Peak-RSS entries report the median paired restart-block ratio with a deterministic bootstrap 95% interval, followed after the semicolon by the raw ratio of observed maxima without an interval.

Artifact	Category	Pairs/blocks	Exact matches (M4/A18)	Latency ratio [95% CI]	Block peak-RSS ratio [95% CI]; raw max
0.5B Q4	simple	400/20	268/268	5.767 [5.683,5.838]	0.972 [0.963,0.984]; 0.995
1.5B Q4	simple	400/20	332/332	9.329 [9.288,9.374]	0.920 [0.914,0.939]; 0.966
0.5B Q4	multiple	200/10	130/130	7.008 [6.554,7.405]	1.003 [0.952,1.011]; 1.011
1.5B Q4	multiple	200/10	155/155	6.441 [6.288,6.636]	0.886 [0.874,0.897]; 0.892
0.5B Q4	parallel	200/10	42/42	4.413 [4.323,4.490]	0.998 [0.938,1.004]; 1.011
1.5B Q4	parallel	200/10	104/104	5.823 [5.740,5.892]	0.877 [0.873,0.893]; 0.870
0.5B Q4	parallel-multiple	200/10	19/19	5.062 [5.007,5.127]	0.971 [0.938,1.000]; 0.969
1.5B Q4	parallel-multiple	200/10	77/77	5.894 [5.799,6.021]	0.855 [0.826,0.871]; 0.859
0.5B Q4	irrelevance	240/12	240/240	6.071 [5.750,6.485]	1.008 [0.969,1.010]; 1.012
1.5B Q4	irrelevance	240/12	211/211	8.310 [8.154,8.455]	0.963 [0.953,0.978]; 0.989

Matched cross-host transfer

A18 Pro host divided by M4 Max - restart-every-20 - 0/2,480 exact-match disagreements - 100.0% raw-output agreement

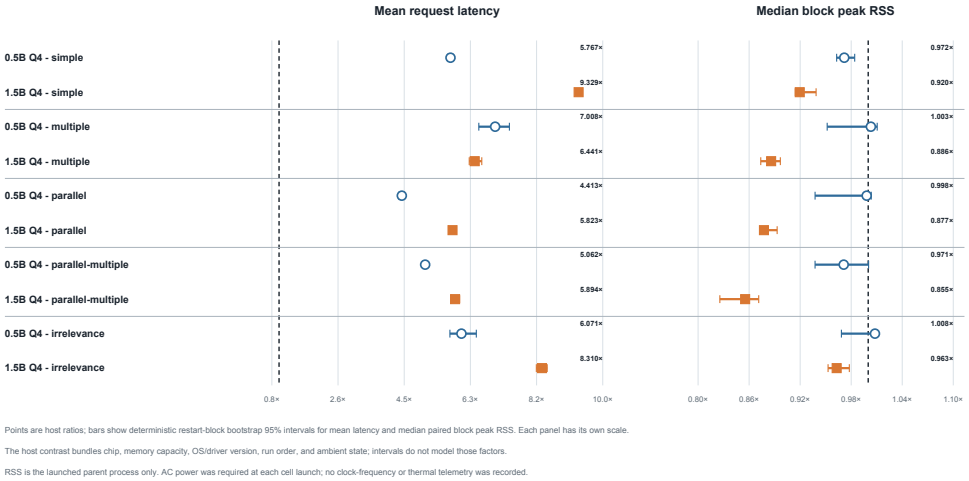


Fig. 7. Matched host-system transfer. Points are A18-Pro/M4-Max ratios; bars are deterministic restart-block bootstrap 95% intervals for mean request latency and median paired block peak parent-process RSS. Raw ratios of observed RSS maxima appear separately in Table 15. The panels use different scales.

corroborating evidence. Resource behavior does not collapse to the same conclusion: A18-Pro/M4-Max mean request-latency ratios range from 4.413 to 9.329 (median 5.982). The largest ratio is 2.11 times the smallest, so even under one binary there is no single aggregate request-time multiplier for these category files and artifacts.

Each ratio is computed within one matched artifact–category cell, not after pooling workloads. Because prompts and raw outputs are identical within every pair, dividing each cell’s total latency by its matched total output-token count would leave the A18-Pro/M4-Max ratio algebraically unchanged. The retained client records do not separate server-side prompt evaluation from decoding,

so the data cannot determine whether prefill and decode throughput ratios are more stable; our claim is about complete request latency, not token-generation throughput.

Median paired block peak-RSS ratios split cleanly by artifact. The five 0.5B values are 0.972, 1.003, 0.998, 0.971, and 1.008 (all at least 0.971), whereas the 1.5B values are 0.920, 0.886, 0.877, 0.855, and 0.963 (all at most 0.963), with no overlap. Descriptively, A18 parent-process RSS is near parity for the smaller artifact and 4–15% lower for the larger one. This artifact-specific allocation signature is consistent with host memory pressure or allocator/OS behavior, but the parent-process scope and single host-run order do not identify a mechanism. Across all contrast cells, median block ratios span 0.855–1.008 (median 0.967); raw observed-maximum ratios span 0.859–1.012 (median 0.979) and receive no interval.

10 DISCUSSION

10.1 Local SLM quality belongs to a system

The pinned label-blind matrix shows the 1.5B Qwen artifact ahead by 23.5–45.4 points on all five category files. Changing the declared instruction and requested JSON shape is sufficient to move non-irrelevance exact match by 20.0 points on 0.5B simple, 16.0 on 0.5B multiple, and 10.5 on 1.5B parallel without changing the build, artifacts, cases, evaluator, or lifecycle. Conditioned irrelevance is a manipulation check, not autonomous abstention; the useful result is that the artifact ranking reverses from a –12.1-point conditioned gap to a +45.4-point label-blind gap. More broadly, the request protocol widens the artifact gap on simple and multiple and narrows it on both parallel files. This request-protocol sensitivity on simple and multiple is concentrated in the 0.5B artifact, a bounded two-artifact pattern rather than a general size law. The GLM/Llama and matched LoRA controls further show why family, parameter count, quantization, or adaptation labels cannot summarize deployment quality without a protocol-matched comparison.

10.2 Output transfer does not imply cost transfer

The matched host control separates two meanings of transfer. Every raw output and exact-match outcome recurs on both hosts, supporting byte-level reproducibility of the pinned deterministic protocol for these 2,480 pairs. Request timing does not: mean request latency on the A18 Pro is 4.41 to 9.33 times that on the M4 Max, even though its measured median block and raw-maximum parent-process-RSS ratios both round to $0.86\text{--}1.01\times$ the M4 Max value. A model-quality result can therefore transfer while its deployment-cost numbers cannot be carried across hosts unchanged, and process RSS alone does not predict speed. The largest latency ratio is 2.11 times the smallest under the same binary, directly ruling out one portable aggregate request-time multiplier for this workload. Matched total-output-token normalization leaves the per-cell ratios unchanged; prefill/decode decomposition was not retained. The non-overlapping RSS strata by artifact further show that the host ratio depends on what is loaded, not only which hosts are named.

10.3 Schemas are semantic and physical resources

Schema exposure changes candidate content, token load, resident memory, and quality. A retriever can make a prompt smaller while also moving evidence earlier in context. The historical controls show that order alone can move exact match substantially; the prospective transfer test rejects a uniform benefit but, given its interval width and low 0.5B baselines, does not establish equivalence in the remaining cells.

10.4 Endpoint choice changes the lifecycle verdict

Restart-every-20 bounds process growth but repeatedly pays model startup. A request-latency-only comparison favors restart in these studies, while an end-to-end comparison favors persistence. Parent-process RSS favors restart as an OS-residency endpoint, but the runs do not distinguish retained file-backed pages from anonymous allocation demand. These are not conflicting measurements to average into one score; they answer different deployment questions.

11 VALIDITY AND LIMITATIONS

External validity. Most measurements use one M4 Max; only the matched ten-contrast-cell Qwen subset repeats on the MacBook Neo with A18 Pro. Two Apple hosts improve the hardware boundary but do not establish behavior on other Apple chips, mobile devices, servers, languages, or production workloads. Both hosts use one primary server implementation and a small set of local artifacts. Both lifecycle studies remain M4-Max-only; whether their persistence tradeoff changes under the MacBook Neo’s 8 GiB memory constraint remains an important follow-up.

Workload validity. The cases are complete local conversions of five BFCL-v3 category files, not an official leaderboard run. Exact tool-call match does not measure tool execution, downstream utility, planning, multi-turn recovery, or every aspect of agent behavior. Accordingly, “SLM agent” in this paper refers specifically to the measured local tool-use stack. Cardinality-conditioned exact match also does not measure autonomous relevance or call-count judgment. Label-blind exact match jointly measures that judgment, JSON-format compliance, and correct call generation; it is not a pure decision-quality metric.

Artifact validity. Model comparisons are artifact comparisons. Architecture, pretraining, instruction tuning, size, and quantization can differ. Earlier Qwen cells lack an exact server-build record and retained artifact SHA-256 values; named paths cannot establish byte identity across study epochs. Newer controls pin their executables and artifacts. MLX and GGUF measurements are not cross-runtime performance comparisons.

Measurement validity. Parent-process RSS excludes other processes, mixes resident file-backed pages with anonymous allocations, and cannot be read as total unified memory or memory demand. Request latency is not decode-only latency; the retained cross-host records cannot separate prompt evaluation from decoding. Original artifact and route cells contain insufficient randomized repeats for causal speed ranking, so those latencies remain quarantined. Lifecycle runs did not record clock frequency or thermal pressure, so their request-latency mechanism is unresolved. The ordered direct-energy traces are retained in the ancillary audit but removed from evidentiary tables and claims because they lack idle subtraction, counterbalancing, and replication. Cross-host runs record AC power only at cell launch and omit clock frequency, thermal pressure, and whole-system power. Automatic GPU offload was inherited from the build default; full Metal offload is verified by later startup probes rather than retained trial-time layer logs. Prompt arms ran sequentially rather than interleaved, so their latency and RSS differences also include run-order and ambient-state uncertainty. Their 96-token ceiling is frequently reached on multi-call categories, so those exact-match effects include template-by-output-limit interaction and are not truncation-free estimates.

Statistical validity. Intervals condition on fixed category files and do not model benchmark construction. Schema seeds are blocked perturbations around 40 source queries, not independent tasks. The historical clustered analysis was finalized after some outcomes were visible; only the newer

order study was prospectively frozen in its final form. The prospective order study declared neither an equivalence margin nor a power target; non-significant contrasts with wide intervals are not null or equivalence findings. The request-treatment protocol and analysis family were frozen after the conditioned comparator was known but before label-blind outcomes; the treatment is prospective, while the full two-arm study is not preregistered. The arms have one run each and fixed order. The host-transfer resampling summaries and compact audit were also finalized after outcomes were available. The median block-RSS bootstrap avoids treating a sample maximum as a regular bootstrap estimand; raw observed maxima are descriptive and receive no interval. Intervals condition on the 10–20 executed restart blocks reported per contrast cell in Table 15 and do not model repeated host runs. Observed zero lifecycle differences and zero cross-host output differences do not establish general equivalence.

Identification validity. The base route grid bundles content and order. Only identical-multiset contrasts identify order. Width contrasts change token count and schema content together. The ancillary energy capture always runs top-8 before top-16, with one trace per execution condition and no idle subtraction, so it licenses no width or efficiency claim. The host transfer bundles chip, memory capacity, OS/driver version, run order, and ambient state; it does not identify a chip-only causal effect. The pinned M4 request-protocol contrast removes build, artifact, decoding-setting, and case differences, but its treatment bundles cardinality disclosure, decision wording, and requested JSON shape. It therefore identifies the declared request protocol, not the mechanism of one clause. A stronger mechanism study would counterbalance repeated blocks of templates that differ only in the disclosure clause. The A18 Pro lifecycle extension remains the next systems-transfer follow-up. The pre-checklist 256-token epoch cannot identify a cap effect because it also differs in *O*, has unavailable *A* provenance, and lacks matched *M* byte identity. Under a cap-only rerun of the pinned deterministic protocol, raising the ceiling could affect correctness only for attempts that reached 96 tokens. Table 9 therefore gives a distribution-free worst-case bound on absolute per-arm exact-match change: 23.0–41.0 points conditioned and 19.5–33.0 label-blind on parallel/parallel-multiple, no more than 0.8 on simple/multiple, and less than 1.3 for irrelevance. For the significant 0.5B simple and multiple contrasts, the summed arm bounds are less than 1.3 and at most 1.0 points; the 1.5B parallel bound remains loose. This assumes every cap-hit can flip, recovers no ungenerated continuation, and cannot replace a frozen counterbalanced 96-versus-256 control. Both that control and lifecycle RSS decomposition require new frozen protocols rather than retrospective relabeling.

12 REPRODUCIBILITY

Artifact availability. The versioned evidence and reproducibility package is archived as version 1.1.0 at <https://doi.org/10.5281/zenodo.21863599> under CC BY 4.0. Within that record, the compact, privacy-checked evidence bundle is distributed as `evidence/single-m4-evidence-v1.tar.gz`.

It contains baseline summaries, frozen protocols, experiment-time snapshots of shared measurement files that later evolved, primary/independent reports, 6,400 order observations, 32 lifecycle pairs, 2,480 compact prompt-conditioning pairs with schema-validity, completion-token, and request-latency diagnostics, 2,480 compact cross-host outcome/hash pairs, restart-block metrics, the frozen prompt protocol, the retrospective Metal-offload probe record, and standard-library checkers. Its manifest records every size and SHA-256; weights, benchmark downloads, raw model outputs, logs, and local paths are excluded. Consequently, the public package independently verifies outcome hashes and reported arithmetic but cannot rescore retained generations. It includes the evaluator and runner needed to regenerate them when the separately identified model and case files are available.

From a clean extraction, the checker verifies all hashes; recomputes the order analysis, lifecycle summaries, prompt-conditioning outcomes and tests, and matched host outcomes and ratios; compares admitted reports; and checks the manuscript tables. It does not rerun inference. The local release retains full logs and inputs; release checks regenerate artifacts, enforce claim boundaries, and compare two clean compiles by text and rendered page.

Generative-AI assistance. OpenAI Codex and Anthropic Claude assisted with implementation, manuscript editing, validation, and critique. The human author set the scope and frozen protocols, directed execution, and accepts responsibility for the manuscript, analyses, code, and artifacts.

13 CONCLUSION

Local SLM agent evaluation is a systems problem. In this study, artifact choice changes exact match; schema exposure changes quality, tokens, and resident memory; and serving lifecycle changes resident-memory and timing endpoints without changing paired outputs. Three findings carry the paper. Under one pinned M4 stack, changing the declared instruction and requested JSON shape moves non-irrelevance exact match by up to 20.0 points and reverses which Qwen artifact is better when abstention must be decided rather than supplied. A historically positive schema-order pattern survives familywise correction in only one of sixteen prospectively frozen parallel-category contrasts, not supporting a universal benefit while leaving smaller effects unresolved. One pinned binary and artifact set then produces 2,480/2,480 byte-identical outputs across two Apple hosts while mean request latency on the A18 Pro is 4.41 to 9.33 times that on the M4 Max. Label-blind and cardinality-conditioned scores answer different questions, and output transfer does not imply cost transfer. The durable result is a measurement principle, operationalized by the minimum record in Section 3.4: report the host, artifact, context construction, evaluator, endpoint scope, and process lifecycle as one declared protocol, then test transfer rather than assuming it.

A PRE-CHECKLIST ARTIFACT MATRIX

Table 16. Pre-checklist label-blind Qwen Q4 artifact matrix; exact server build unavailable. Differences are 1.5B minus 0.5B on paired case IDs with 95% intervals; RSS is sampled peak process RSS. Holm correction covers five McNemar tests.

Category	Artifact	Exact	Paired difference (pp)	Holm p	Peak RSS (MiB)
simple	Qwen2.5 0.5B Q4	269/400			637.4
	Qwen2.5 1.5B Q4	339/400	+17.5 [12.8, 22.2]	6.69×10^{-12}	1335.0
multiple	Qwen2.5 0.5B Q4	122/200			689.7
	Qwen2.5 1.5B Q4	153/200	+15.5 [8.6, 22.4]	3.31×10^{-5}	1423.5
parallel	Qwen2.5 0.5B Q4	63/200			714.4
	Qwen2.5 1.5B Q4	109/200	+23.0 [14.3, 31.7]	3.75×10^{-6}	1412.7
parallel-multiple	Qwen2.5 0.5B Q4	59/200			752.5
	Qwen2.5 1.5B Q4	98/200	+19.5 [11.6, 27.4]	1.05×10^{-5}	1531.0
irrelevance	Qwen2.5 0.5B Q4	31/240			631.3
	Qwen2.5 1.5B Q4	119/240	+36.7 [30.5, 42.9]	7.35×10^{-25}	1304.4

This matrix is retained to document the pre-checklist measurement epoch, not to anchor the paper’s artifact claim. Its category files and evaluator are known, but the exact server build and artifact SHA-256 values were not retained. The direction agrees with the pinned label-blind comparison; the different magnitude illustrates why numerical values should not be transported across protocol epochs.

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